BY IMRAN SSETTUMBA

GROUP 7



LAYER 7 APPLICTION LAYER

GROUP 7 MEMBERS

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Overview of layer 7

The topmost layer

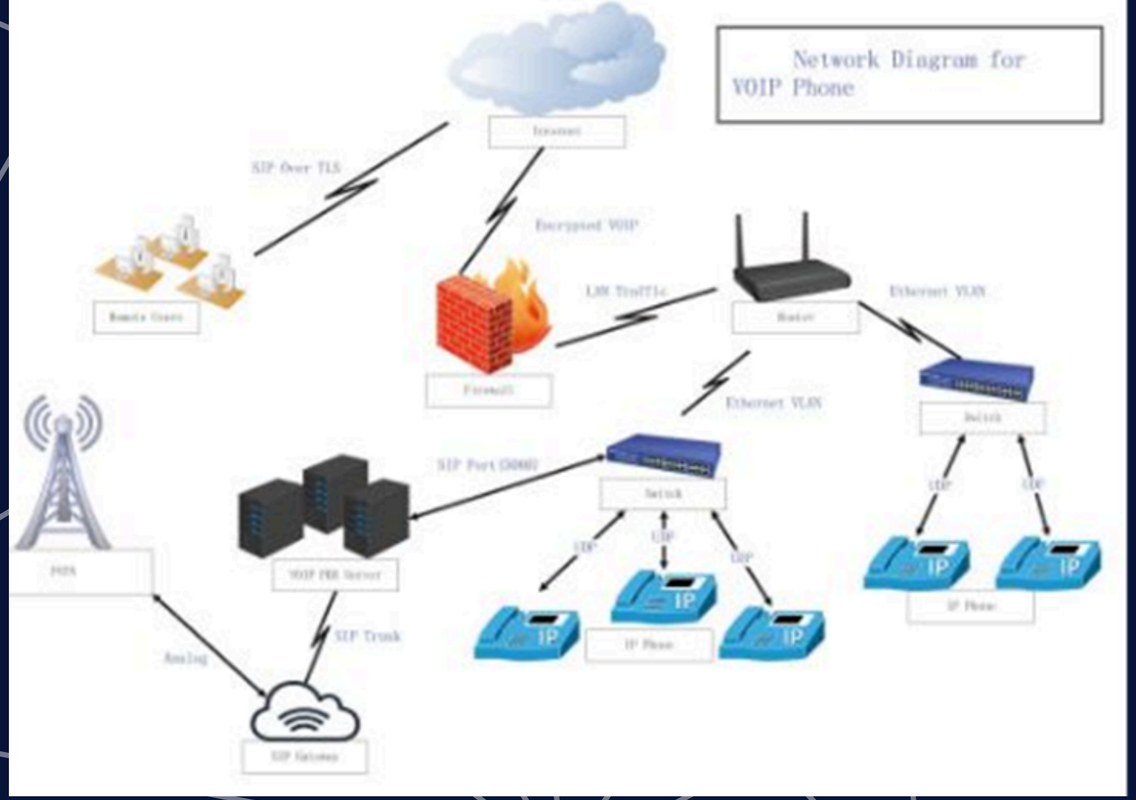


The Application Layer serves as the direct interface for end-user applications to access network services. Unlike software programs themselves, Layer 7 provides the standardized protocols that enable software to send and receive data across networks. It translates user requests into protocol-compliant network transmissions, ensuring inter-operability across disparate operating systems and device hardware.

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FUNCTIONS

- Partner Identification: Determines the availability and identity of communication partners for a intended data exchange.
- Resource Availability: Checks whether sufficient network and system resources exist to establish communication.
- Communication Sync: Coordinates software requests and synchronizes data exchange between endpoints
- Service Abstraction: Provides high-level APIs so application code does not need to handle underlying transport mechanics.



Smart Infrastruct

Digital Innovation

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NextWave Connectivity

Key Application protocols

HTTP

hypertext transfer Protocal gavens web communications. HTTPS Adds SSL/TLS encryption for secure web browsing and

SMTP/IMAP

Simple Mail TransferProtocol handles outbound email routing,while MAP/POP3 manages inboundmailbox access and folder sync.

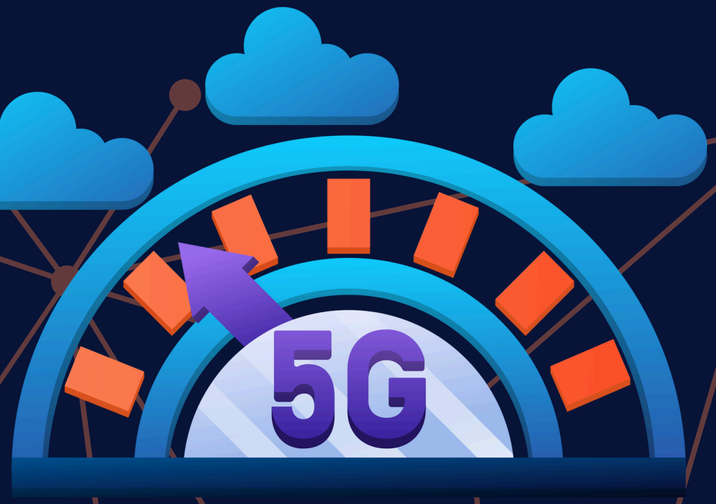
FTP/SFTP

File Transfer Protocolprovides robustclient-server file upload and download capabilities, enhanced by SFTP over SSH

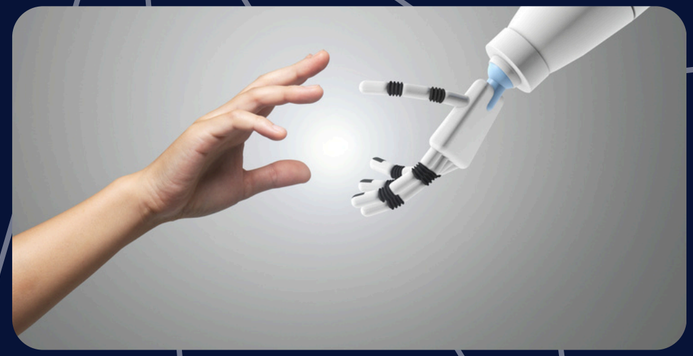
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NextWave Connectivity

DNS & Name Resolution







The Internet Directory

Domain Name System DNS) operates at Layer 7 to convert human-friendly hostnames (e.g., example.com) into machine-routable IP addresses.

It employs a distributed hierarchical database consisting of Root, Top-Level Domain TLD□, and Authoritative nameservers, utilizing UDP port 53 for lightweight queries.

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NextWave Connectivity

Architectural models



Client-Server Model

Centralized model where designated server hosts resources and listens for incoming requests.Clients initiate communication to retrieve data or execute procedures.

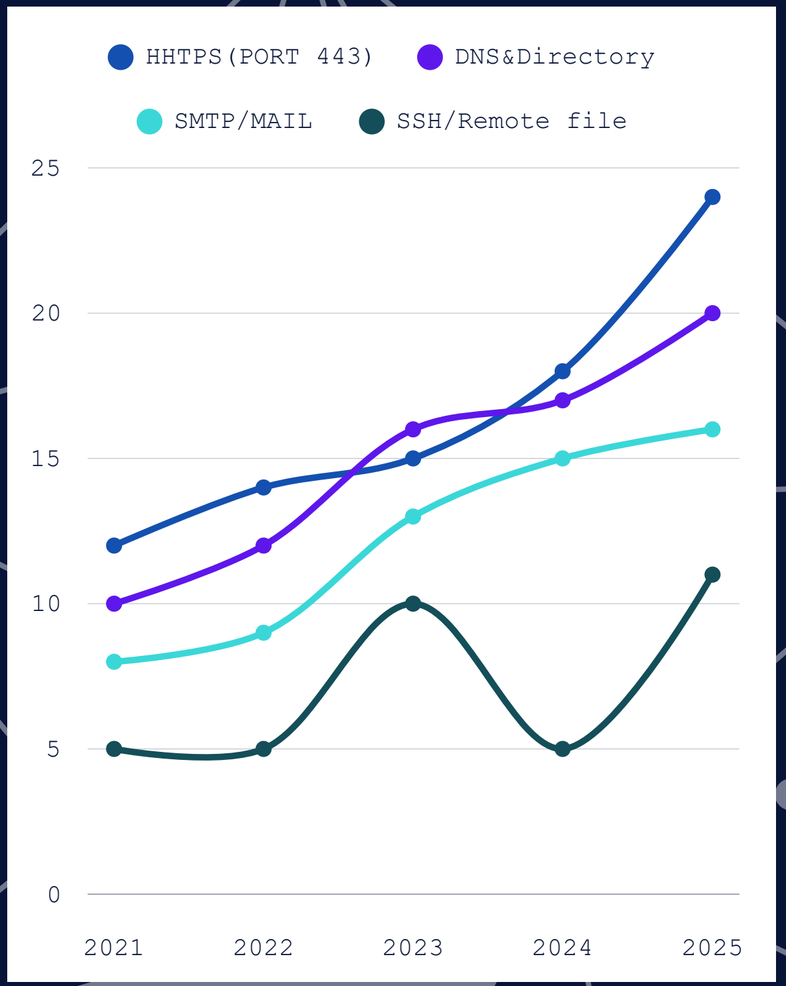
Peer-to-Peer (P2P) Model

Decentralized model where nodes act as both resource consumers and providers. Eliminates single points of failure and scales dynamically with peer volume.
Examples: BitTorrent file distribution, Blockchain networks, InterPlanetary File System IPFS.

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NextWave Connectivity

Layer 7 Protocol traffic share



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Protocol	Transport Type	Standard Port	Primary Role
HTTP / HTTPS	TCP	80 / 443	Web document delivery & API data transfer
DNS	UDP	53	Domain name to IP address translation
DHCP	TCP		Automated client IP configuration
SSH / SFTP	TCP	22	Secure remote CLI administration & file sync
SNMP	UDP	161 / 162	Network device monitoring and metrics collection

NextWave Connectivity

70% layer 7 security

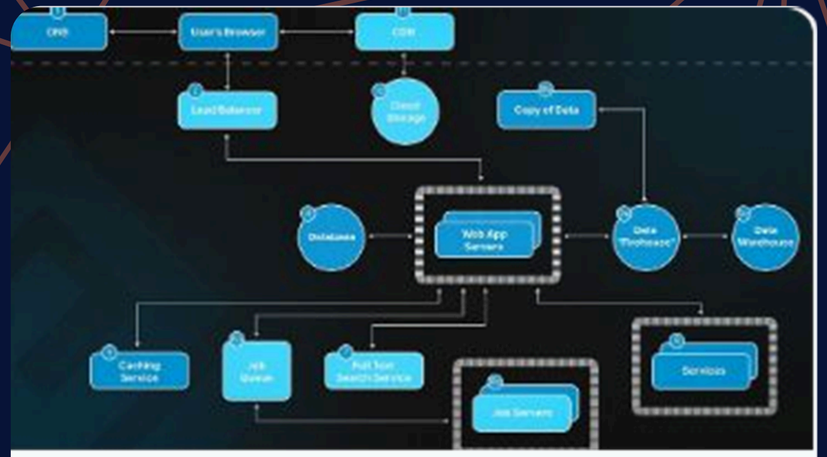


Targeting Application Vulnerabilities

Modern cyber threats have shifted upwards from network layer packets to Layer 7 payload exploits. Distributed Denial of Service DDoS) and API abuse increasingly target application endpoints directly. Deploying Web Application Firewalls WAF) at Layer 7 allows organizations to inspect HTTP payloads and block malicious SQL injections, XSS, and bot traffic before reaching backend systems.

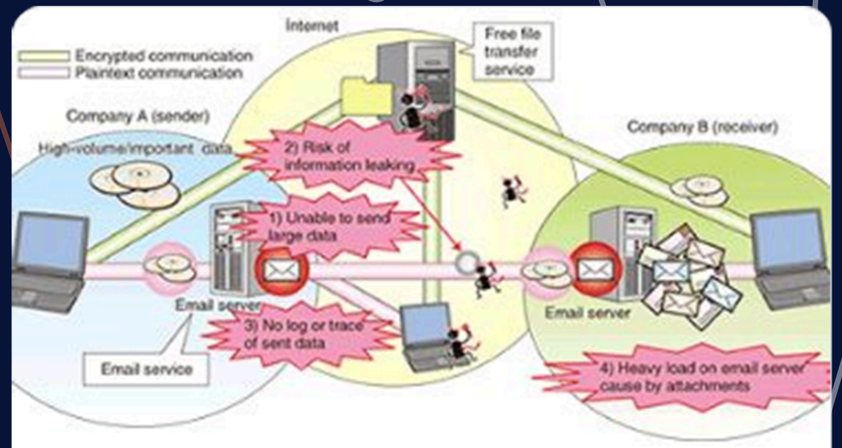
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Real-World Application



Web Browsing & SaaS

Web browsers use HTTP/3 and WebSocket protocols at Layer 7 to render interactive SaaS applications.



Enterprise Messaging

Email platforms coordinate SMTP relays and IMAP synchronization to deliver global electronic mail.



Remote Management

Sysadmins rely on SSH and SNMP protoco remotely execute commands and manage fleets.

Evolution of Network Layer 7

1984

ISO standardizes the 7‑Layer OSI Reference Model.

1990s

HTTP/1.0 and DNS revolutionize world wide web growth.

2010s

HTTPS encryption becomes mandatory across major browsers.

Present

HTTP/3 over QUIC accelerates clou APIs and edge services.

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Thank You

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